



MARKET MONITOR

No. 44 – December 2016

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Roundup

As the year draws to a close, the positive supply situation that has characterized global markets for the four AMIS crops in recent years seems to extend into 2017. Assuming no major production setbacks occur, prices in international markets are thus likely to remain stable in the coming months, although external factors such as currency movements and developments in energy markets remain sources of uncertainty.

Markets at a glance

	From previous forecast	From previous season
Wheat	▲	▲
Maize	▲	▼
Rice	■	■
Soybeans	▲	■

▲ Easing ■ Neutral ▼ Tightening

The **Market Monitor** is a product of the Agricultural Market Information System (AMIS). It covers the international markets for wheat, maize, rice and soybeans, giving a synopsis of major market developments and the policy and other market drivers behind them. The analysis is a collective assessment of the market situation and outlook by the eleven international organizations and entities that form the AMIS Secretariat.

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World supply-demand outlook

- **Wheat** production in 2016 raised, reflecting upward revisions in Kazakhstan, Iran and Nepal.
- Utilization in 2016/17 to expand by 2.8 percent with feed use increasing by 7.2 percent on large supplies and low prices.
- Trade forecast for 2016/17 (July/June) unchanged but this month's export forecasts for Australia and the EU are lifted, offsetting a cut in the forecast for Russia.
- Stocks (ending in 2017) scaled up as end-season inventories in the CIS and Iran are projected higher than anticipated earlier.

WHEAT

Production
Supply
Utilization
Trade
Stocks

FAO-AMIS		
2015/16	2016/17	
est.	f'cast	
	2-Nov	8-Dec
735	747	749
947	972	976
714	733	734
168	168	168
227	235	239

in million tonnes

USDA		IGC	
2015/16	2016/17	2015/16	2016/17
est.	f'cast	est.	f'cast
	8-Nov		24-Nov
735	745	737	749
953	986	942	970
712	736	720	736
172	174	164	166
241	249	222	235

- **Maize** production in 2016 increased following a significant upward revision in the US, more than offsetting a downward revision in India.
- Utilization in 2016/17 to increase by 2.4 percent, driven by significant expansions in China and the US for animal feed.
- Trade in 2016/17 (July/June) to contract, mainly on reduced import demand in China and several countries in Central America.
- Stocks (ending in 2017) scaled up mainly on expectation of a record build-up of inventories in the US.

MAIZE

Production
Supply
Utilization
Trade
Stocks

FAO-AMIS		
2015/16	2016/17	
est.	f'cast	
	2-Nov	8-Dec
1006	1025	1027
1229	1240	1244
1007	1030	1030
139	137	137
217	208	212

in million tonnes

USDA		IGC	
2015/16	2016/17	2015/16	2016/17
est.	f'cast	est.	f'cast
	8-Nov		24-Nov
959	1031	971	1042
1167	1240	1178	1250
959	1022	970	1026
120	144	136	135
209	218	208	224

- **Rice** production in 2016 upgraded on more upbeat area expectations, especially for Bangladesh and Mali.
- Utilization in 2016/17 little changed from November, as prospects of higher intake in West Africa outweigh downward revisions for India and China.
- Trade in calendar 2017 downscaled, primarily reflecting lower anticipated cross-border purchases by China.
- Stocks (ending in 2017) seen broadly steady y/y, following upward revisions to inventories held by Myanmar and Viet Nam.

RICE (milled)

Production
Supply
Utilization
Trade
Stocks

FAO-AMIS		
2015/16	2016/17	
est.	f'cast	
	2-Nov	8-Dec
491	498	499
666	669	670
495	501	501
42.7	43.4	42.9
171	170	171

in million tonnes

USDA		IGC	
2015/16	2016/17	2015/16	2016/17
est.	f'cast	est.	f'cast
	8-Nov		24-Nov
472	484	472	485
587	600	590	601
470	478	474	483
39.7	41.0	40	41
116	122	116	118

- **Soybean** production in 2016/17 lifted by another 3 million tonnes owing to record-high yields in the US and small upward revisions for the EU, Canada and China.
- Utilization in 2016/17 to expand by about 5 percent y/y, with world consumption falling below global production.
- Trade in 2016/17 scaled up slightly on higher than anticipated import demand by China and the EU, while export forecasts are raised for the US and Canada.
- Stocks forecast (2016/17 carry-out) increased by 3.6 million tonnes or 9 percent, reflecting higher estimates for the US, Brazil and Argentina. Global inventories now expected to match the 2015/16 all-time high.

SOYBEANS

Production
Supply
Utilization
Trade
Stocks

FAO-AMIS		
2015/16	2016/17	
est.	f'cast	
	2-Nov	8-Dec
314	333	336
358	375	379
320	334	335
135	138	139
43	39	43

in million tonnes

USDA		IGC	
2015/16	2016/17	2015/16	2016/17
est.	f'cast	est.	f'cast
	8-Nov		24-Nov
313	336	315	336
392	413	353	370
316	329	319	332
132	139	133	137
77	82	34	38



FAO-AMIS monthly forecast

For latest revisions to FAO-AMIS monthly forecasts for 2016/17 see next page.

To review and compare data, by country and commodity, across the three main sources, go to:

<http://statistics.amis-outlook.org/data/index.html#COMPARE>

Summary of revisions to FAO-AMIS monthly forecasts for 2016/17

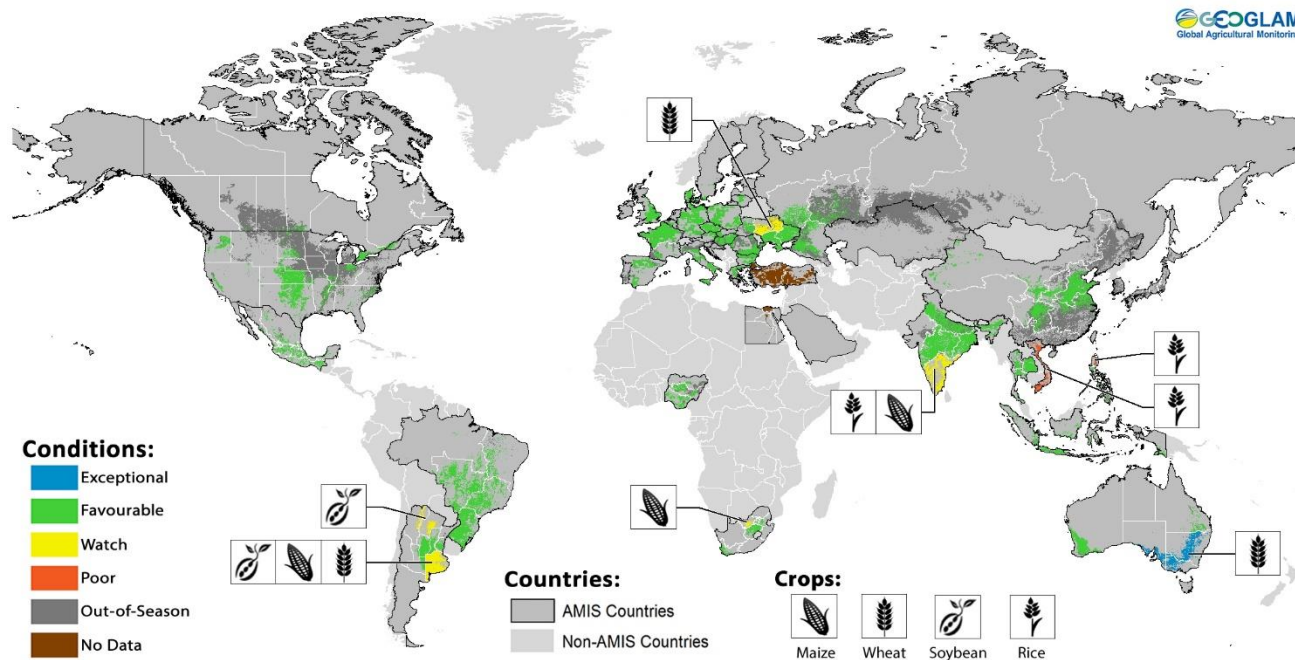
in thousand tonnes

	WHEAT					MAIZE				
	Production	Imports	Utilization	Exports	Stocks	Production	Imports	Utilization	Exports	Stocks
WORLD	2631	530	844	505	3254	2874	992	339	932	4064
Total AMIS	983	-150	-444	45	1736	3169	700	866	570	2862
Argentina	-	-	-	-	-	-	-	-	-	-
Australia	-	-	-	500	-	-	-	-	-	-
Brazil	92	-	-8	-	150	-267	-	-267	-	-
Canada	-	-	314	-	-	-	-	-	-	-
China Mainland	-	200	-	-50	200	-	-	-	-	-
Egypt	-	200	200	-	-	-	200	200	-	-
EU	-	-	-200	500	-	-	-	-	-	-
India	-	500	300	-50	-	-1000	-150	-	-	-1184
Indonesia	-	-600	190	-	-200	200	100	200	-200	200
Japan	-	-	-	-	-	-	-	-	-	-
Kazakhstan	900	-	100	-	800	-	-	-	-	-
Mexico	-	200	100	-	100	-	-	-1300	500	1500
Nigeria	-	-	-	-	-	-135	-	-135	-	-
Philippines	-	200	100	-	100	-	-	-	-	-
Rep. of Korea	-9	600	141	-	200	-5	-	-195	-	103
Russian Fed.	-	-	-	-1000	1000	-	-	-	100	-100
Saudi Arabia	-	-	-	-	-	-	-	-	-	-
South Africa	-	-	-	-	-	-	-	-	-	-
Thailand	-	300	200	-	100	430	-	30	170	230
Turkey	-	-	-	-	-	-	-	-	-	-
Ukraine	-	-1950	-1700	-	-950	-	-	-	-	-
US	-	-	-136	-	136	4272	50	2159	-	2113
Viet Nam	-	200	-45	145	100	-326	500	174	-	-

	RICE					SOYBEANS				
	Production	Imports	Utilization	Exports	Stocks	Production	Imports	Utilization	Exports	Stocks
WORLD	599	-530	-41	-489	779	2907	1020	788	902	3618
Total AMIS	-6	-420	-531	-100	-4	2907	1570	1348	902	3733
Argentina	-	-	2	-	-	-	-	-100	-	300
Australia	-	-	-	-	-	-	-	-	-	-
Brazil	-	-50	-	-	-	-388	-10	-10	-300	812
Canada	-	-	-	-	-	224	-	-1	200	25
China Mainland	-	-400	-500	-	-	200	1000	1200	-	-
Egypt	-	-	-	-	-	-	-	-	-	-
EU	25	-10	-5	-	30	364	337	764	-	-
India	-	-	-300	200	-	-	-	-	-	-
Indonesia	-	-	-	-	-	-	-	-	-	-
Japan	22	-	2	-	20	11	393	304	-	224
Kazakhstan	-	-	-5	-	5	-	-	-	-	-
Mexico	-3	-	2	-	-	-	-50	-200	-	100
Nigeria	-	-	-20	-	-	-	-	-	-	-
Philippines	-	-	-	-	-	-	-	-	-	-
Rep. of Korea	-5	40	75	-	-101	-	-	-	-	-
Russian Fed.	-	-	-	-	-	-	-	-	-	-
Saudi Arabia	-	-	-10	-	-60	-	-100	-100	-	-
South Africa	-	-	-	-	-	-8	50	2	2	-12
Thailand	-	-	-	-	-	-	-50	-20	-	-30
Turkey	-6	-	-1	-	-10	-	-	-	-	-
Ukraine	-	-	-	-	-	-	-	-	-	-
US	-39	-	-1	-	-38	2504	-	-491	1000	2314
Viet Nam	-	-	230	-300	150	-	-	-	-	-

Crop monitor

Crop conditions in AMIS countries (as of 28 November)



Crop condition map synthesizing information for all four AMIS crops as of 28 November. Crop conditions over the main growing areas for wheat, maize, rice, and soybean are based on a combination of national and regional crop analyst inputs along with earth observation data. **Only crops that are in other-than-favourable conditions are displayed on the map with their crop symbol.**

La Niña update

La Niña conditions are established in the equatorial Pacific Ocean. They are anticipated to persist through January 2017, followed by transition to a neutral state. Consistent with this, broad areas of the Horn of Africa are experiencing a drier than normal October-December rainy season, with some areas characterized by drought with the worst vegetation conditions in fifteen years. Drier-than-normal conditions are expected in the next three months in southwest Asia, southeastern China, southeastern South America, and the southern United States. Above average rainfall is favored for southern Africa, Southeast Asia, Australia, and northern South America.

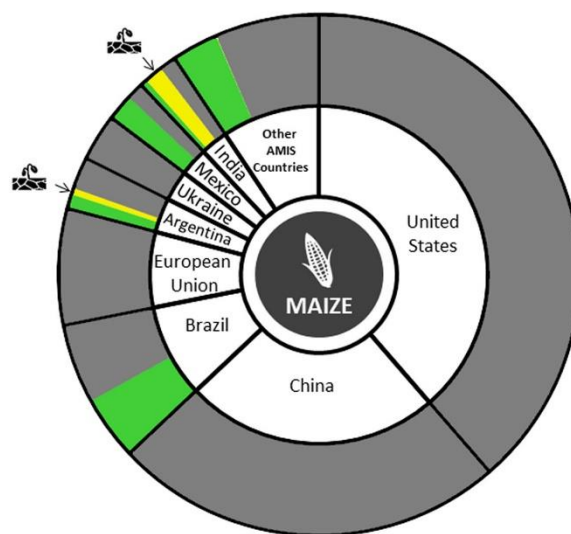
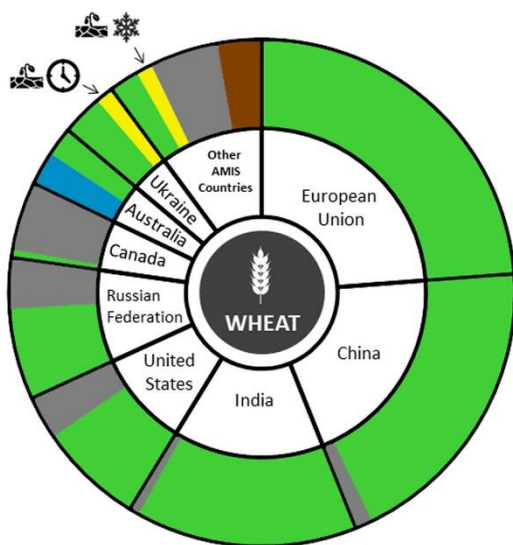
Conditions at a glance

Wheat - In the northern hemisphere, winter wheat planting is complete and is under generally favourable conditions at this early stage in the season as it enters dormancy in most countries. In the southern hemisphere, harvest has begun with conditions mostly favourable, for South Africa, and most of Argentina. Australia is experiencing exceptional conditions over large production areas, owing to earlier beneficial rainfall.

Maize - In the northern hemisphere, harvest is ongoing in India under generally favourable conditions, with the exception of the south, due to rainfall deficits. Conditions are favourable in Canada and Mexico. In the southern hemisphere, conditions for planting are generally favorable for Argentina, Brazil, and South Africa.

Rice - Rice conditions for Southeast Asia are mixed. In southern India, where the Kharif crop harvest is ongoing, there is concern due to rainfall deficits. Conditions are favourable in Indonesia, Thailand, and most of the Philippines. By contrast, conditions are poor in Viet Nam due to a mix of adverse weather conditions during the growing season.

Soybeans - In the northern hemisphere, the season is largely complete. In the southern hemisphere, conditions in Brazil are favourable and planting is underway in Argentina under favourable condition in the central agricultural region. However, dry soil moisture conditions in the northern and southern areas has slowed planting.

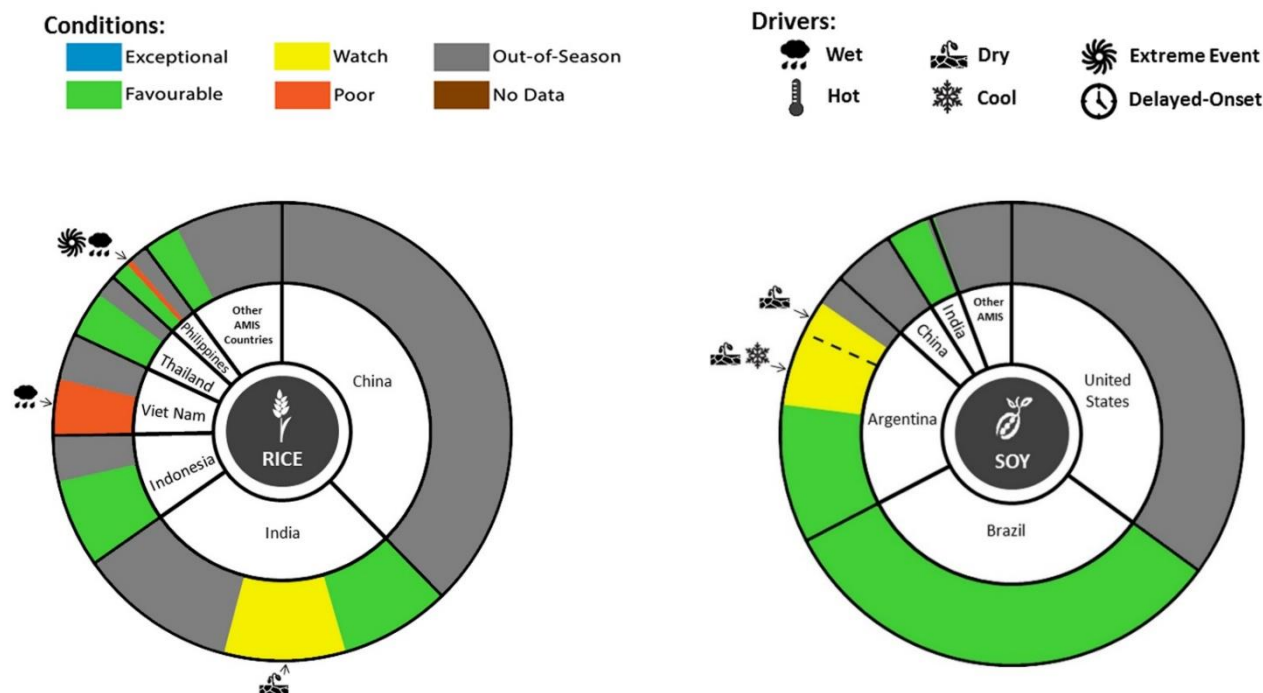


Wheat

In the **EU**, overall conditions are favourable for winter wheat despite some problematic weather conditions during sowing and emergence. In the **US**, conditions are favourable for winter wheat. In **Canada**, winter wheat is under favourable conditions with only minor delays in seeding due to wet conditions in Saskatchewan. In **China**, conditions have improved with recent good weather and beneficial snowfall for the dormancy period. In **India**, sowing is ongoing under generally favourable conditions. In the **Russian Federation**, winter wheat conditions are favourable as the crop enters dormancy. In **Ukraine**, winter wheat is under generally favourable conditions, and November rains provided good soil moisture reserves. However, there is concern over conditions in central Ukraine where seeds in a number of areas did not form shoots due to late planting and cold temperatures in October. In **Australia**, record yields are expected as conditions have improved markedly since September, with abundant soil moisture levels and mild temperatures boosting production prospects across the country. In **Argentina**, harvest is ongoing in the north of the country under favourable conditions. The main producing regions are under generally favourable conditions with good yields expected. However, a mix of dry soils, flooding, and frosts is causing some concern in parts of the Buenos Aires region.

Maize

In **India**, harvest is ongoing under generally favourable conditions. However, in the south there are some concerns due to rainfall deficits. In **Ukraine**, conditions are favourable, however the completion of harvest is delayed in some areas due to the rainy weather. In **Canada**, harvest delays continue due to wet conditions in the central prairies. In Ontario prospects improved owing to late rains, resulting in higher yields than previously expected. In **Mexico**, the spring-summer crop is under favourable conditions and a larger crop than last year is expected. In **Nigeria**, conditions are favourable for the second maize crop. In **Brazil**, conditions for planting and development of the spring crop are generally favourable. In **Argentina**, conditions are favourable with planting of early maize expanding into Buenos Aires and La Pampa regions owing to good soil moisture conditions, with the exception of some areas in the center and south. Late maize planting will begin soon. In **South Africa**, planting is progressing under favourable conditions owing to normal to above-normal spring rainfall.



Rice

In **India**, Kharif crop harvest and land preparations for the Rabi crop are ongoing under generally favourable conditions, except in the south due to dry conditions caused by rainfall deficits over the last 2 months. In **Indonesia**, wet season planting is ongoing under favourable conditions owing to the early onset of the rainy season in September. There is an increase in planted area this season owing to the combination of favourable weather and government support. In **Viet Nam**, harvest of the autumn-winter crop in the north is ongoing and yields are expected to be below average (and similar to last year's drought affected crop) due to a mix of adverse weather conditions during the growing season. In the south, harvest of the summer-autumn crop is also ongoing, with below average yields expected. Land preparations are ongoing for dry season rice. In **Thailand**, the wet season crop is in the grain filling stage and growing conditions are favourable owing to good rainfall. Land preparations ongoing for the dry season crop. In the **Philippines**, wet season rice, planted in July-August, is in maturing to harvesting stages under generally favourable conditions. Crop damage and flooding from Typhoon Haima (Lawin) resulted in poor conditions in the north.

Soybeans

In **Brazil**, planting is complete in the main producing regions and is under favourable conditions. In **Argentina**, favourable weather conditions promoted planting in the center of the agricultural region. Dry weather in the previously flooded areas improved seeding conditions, however combined with low temperatures in the southern parts of Buenos Aires region, it caused some planting delays. In **India**, harvest is completing with favourable prospects owing to good monsoon rains this season.

Information on crop conditions in non-AMIS countries can be found in the [GEOGLAM Early Warning Crop Monitor](#), published 8 December 2016

i Pie chart description: Each slice represents a country's share of total AMIS production (5-year average), with the main producing countries (90 percent of production) shown individually and the remaining 10 percent grouped into the "Other AMIS Countries" category. Sections within each country are weighted by the sub-national production statistics (5-year average) of the respective country and accounts for multiple cropping seasons (i.e. spring and winter wheat).

The late vegetative through to reproductive crop growth stages are generally the most sensitive periods for crop development.

Sources and Disclaimers: The Crop Monitor assessment is conducted by GEOGLAM with inputs from the following partners (in alphabetical order): Argentina (Buenos Aires Grains Exchange, INTA), Asia Rice Countries (AFSIS, ASEAN+3 & Asia RICE), Australia (ABARES & CSIRO), Brazil (CONAB & INPE), Canada (AAFC), China (CAS), EU (EC JRC MARS), Indonesia (LAPAN & MOA), International (CIMMYT, FAO, IFPRI & IRR), Japan (JAXA), Mexico (SIAP), Russian Federation (IKI), South Africa (ARC & GeoTerraImage & SANSA), Thailand (GISTDA & OAE), Ukraine (NASU-NSAU & UHMC), USA (NASA, UMD, USGS – FEWS NET, USDA (FAS, NASS)), Viet nam (VAST & VIMHE-MARD). The findings and conclusions in this joint multiagency report are consensual statements from the GEOGLAM experts, and do not necessarily reflect those of the individual agencies represented by these experts. More detailed information on the GEOGLAM crop assessments is available at www.geoglam-crop-monitor.org

Policy developments

Wheat

- On 4 November, the Ministry of Agriculture of **Brazil** authorized the national supply company (CONAB) to hold wheat auctions in the event of a price fall below BRL 644.17 (USD 205.48) per tonne in order to ensure farmers' revenue. On 17 November, Brazil also announced the provision of further financial support to local wheat sales after a record domestic crop depressed prices and market activity. Around BRL 150 million (USD 43.8 million) will be assigned to helping farmers sell up to 1.7 million tonnes of wheat to the local food industry.
- On 30 October, the Wheat Committee in **Egypt** submitted a recommendation to the Ministry of Agriculture to increase the prices paid to wheat producers from EGP 2,800 (USD 315) per tonne to EGP 3,000 (USD 338) per tonne. This recommendation is pending official government approval.
- On 14 November, the Government of **Egypt** authorized the General Organization for Export and Import Control (GOEIC) to conduct safety inspection procedures for wheat, at both arrival and departure ports. Replacing the previously established quarantine procedures, this measure is expected to simplify the import and export inspection while minimizing trade costs.
- On 4 November, the Government of **India** announced plans to sell one million tonnes of wheat from its strategic buffer stock in the open market, in response to rising wheat prices in several parts of the country. On 15 November, the Indian Cabinet Committee on Economic Affairs also approved an increase in minimum support price for wheat and other Rabi crops in the 2016-2017 season. In the case of wheat, the minimum support price will increase by 6.6 percent, from INR 15,250 (USD 225) per tonne to INR 16,250 (USD 239.76) per tonne.
- On 12 November, during the Fourth Joint Commission meeting between **Nigeria** and **Russian Federation** held in Abuja, the Nigerian Ministry of State for Foreign Affairs disclosed plans to halt wheat imports from Russia due to tight foreign currency reserves. Instead, Nigeria would be seeking greater co-operation from the Russian Federation to foster technology transfer and improve agricultural productivity.

Rice

- On 23 October, the Ministry of Irrigation in **Egypt** decided to reduce the maximum area allowed for the cultivation of rice, a water-intensive commodity, by 34.6 percent. According to the decision, the rice cultivation area in 2017 would be restricted to six governorates, i.e. approximately 704,500 feddans (295,890 hectares).
- On 16 November, the **EU** notified the WTO of its intention to reduce the maximum residue limits (MRLs) of tricyclazole on imported rice from the current 1 mg/kg to 0.01 mg/kg, with effect from June 2017. In its submission to the WTO, the EU specified a 60-day comment period (i.e. until 15 January 2017).
- On 26 October, the Minister of Agriculture of **Indonesia** announced plans to provide subsidies for hybrid rice seeds, to be sown on 5 million hectares of land, with the goal of achieving self-sufficiency in rice production.
- On 8 November, The National Rice Policy and Management Committee in **Thailand** approved new assistance programmes worth USD \$514 million in order to encourage farmers to delay rice sales and stabilize domestic prices. Under this scheme, white paddy producers would receive BHT 10,500 (USD 299) per tonne (of which a loan payment of BHT 7,000, a harvest subsidy of BHT 2,000 and a storage subsidy of BHT 1,500). Producers of Thai Pathum Thani fragrant rice would receive BHT 11,300 (USD 322) per tonne (of which a loan of BHT 7,800, a harvest subsidy of BHT 2,000 and BHT 1,500 for storage). On 18 November, payments for glutinous rice were set to BHT 13,000 (USD 370) per tonne, on par with outlays announced for Hom Mali rice on 1 November 2016. As with the other rice varieties, the outlay would be inclusive of a BHT 2,000 per tonne compensation for quality improvement and harvesting costs and BHT 1,500 as a subsidy for storage costs. This last support measure will remain in place until 28 February 2017.

Soybeans

- On 1 November, **Argentina's** National Health Service and Food Quality Organization (SENASA) and **Russia's** Veterinary and Phytosanitary Federal Service signed an agreement to resume Argentinean exports of soybean meal and flour to Russia and other Eurasian Economic Union members. These had been suspended



AMIS Policy database

Visit the AMIS Policy database at: <http://statistics.amis-outlook.org/policy/>

The **AMIS Policy database** gathers information on trade measures and domestic measures related to the four AMIS crops (wheat, maize, rice, and soybeans) as well as biofuels. The design of this database allows comparisons across countries, across commodities and across policies for selected periods of time.

since February 2016 over concerns about genetically modified organisms (GMOs).

- On 16 November, **Indonesia** raised the farmgate floor prices of soybean to IDR 8,500 per kg (USD 648 per tonne) while setting the consumer ceiling prices at IDR 9,200 per kg (USD 701 per tonne). The reference prices will be revised every four months.

Across the board

- On 16 November, the Ministry of Agriculture of **Brazil** announced a number of trade facilitation measures in the agro-food sector aiming at modernizing the sector and reducing the operating costs incurred by import and export businesses.
- On 24 October, **China** introduced several maximum residue limits on pesticides contained in various commodities, including rice, maize, soybeans and wheat. The deadline indicated by China for receiving comments from the WTO membership is 23 December 2016.
- On 2 November, **Germany** approved draft legislation to ban the cultivation of crops with genetically modified components. The draft legislation was prepared within the EU-wide framework that allows individual member States to opt out of GMO cultivation even if declared safe by the European Food Safety Agency.
- As part of the reform approved in February this year, on 18 November, **India** substituted its crop insurance programme with a new, yield-based scheme. The new scheme will cover yield losses incurred due to risks such as natural disasters, extreme weather conditions, and pest infestation.
- Following the demonetization steps taken by **India**, on 21 November, the National Bank for Agriculture and Rural Development (NABARD) was instructed to disburse INR 210 billion (USD 3.07 billion) to farmers through the network of farm cooperatives in order to ease liquidity conditions. In addition, special measures were taken in favour of farmers in the current Rabi season such as granting higher cash withdrawals, extended time limits for the payment of crop insurance premiums and the purchase of seeds with old denomination bank notes of INR 500 from selected government centres.
- On 17 November, **Nigeria** introduced a number of structural and transparency measures to further the implementation of the Growth Enhancement Scheme, under which rice, wheat, maize and groundnut farmers are eligible for support. The new policies include increasing government participation in the redemption of claims, introducing fertilizer stock tracking, and the

adoption of a single commodity value chain based on comparative advantage.

- On 9 November, the **US** Environmental Protection Agency approved the use of a dicamba herbicide on genetically-modified crops, including soybeans, in order to mitigate the spread of glyphosate-resistant weeds.

Biofuels

- On 30 November, as part of the revision of the Renewable Energy Directive, the **EU** Commission proposed reducing progressively the maximum contribution from liquid biofuels to the renewable energy target from 7.0 percent in 2021 to 3.8 percent in 2030. To this end, reductions will be implemented by 0.3 percentage points annually through 2025; and by 0.4 percentage points for 2026-2030. Simultaneously, the EU Commission proposed replacing conventional biofuels (i.e., crop-based) by second-generation biofuels (i.e., produced from algae, agricultural residues, etc.). The Commission proposal is yet to be approved by individual member States and the EU Parliament.
- For 2017, the **US** Environmental Protection Agency increased the target for total renewable fuel use to 19.28 billion gallons, a six-percent rise from the 2016 target of 18.11 billion gallons. This target includes 15 billion gallons for conventional biofuel and 4.28 billion gallons for the advanced biofuels category which may include soyoil-based biodiesel.

International prices

International Grains Council (IGC) Grains and Oilseeds Index (GOI) and GOI sub-Indices

	Nov 2016 Average*	% Change M/M	% Change Y/Y
GOI	189	- 1.1%	+ 2.7%
Wheat	157	- 1.2%	- 8.7%
Maize	180	- 1.2%	+1.7%
Rice	144	- 1.4%	-4.9%
Soybeans	198	-1.4%	+11.9%

*Jan 2000=100, derived from daily export quotations

Wheat

World wheat export prices continued to be pressured by record supplies, with average values down by about 1 percent m/m at close to 10-year lows. There was some sustained underpinning from concerns about limited availabilities of high quality milling wheat, with price premiums for the best supplies remaining wide. Support also came from less than ideal weather for 2017/18 crops in some areas, including worsening drought for US winter wheat. Nevertheless, at this early stage of the growing season, overall world production prospects for 2017/18 were still seen as broadly favourable, with little change in area thought likely. Amid limited wheat specific news, day-to-day price movements were often linked to other commodities, especially rowcrops, as well as changes in currencies, with the US dollar index at near 14-year highs.

Maize

Average maize export prices eased slightly during November, with the IGC GOI sub-Index down by around 1 percent m/m, weighed by ample feedgrain availabilities and a mostly favourable outlook for the next southern hemisphere crops. US quotations dropped following the near completion of a record harvest, with spot supplies priced at an unusually steep discount to other origins by the end of the month. While declines were partly linked to a drop in CME futures, Gulf premiums were also weak as US exporters looked to secure sales before South American crops are marketed early next

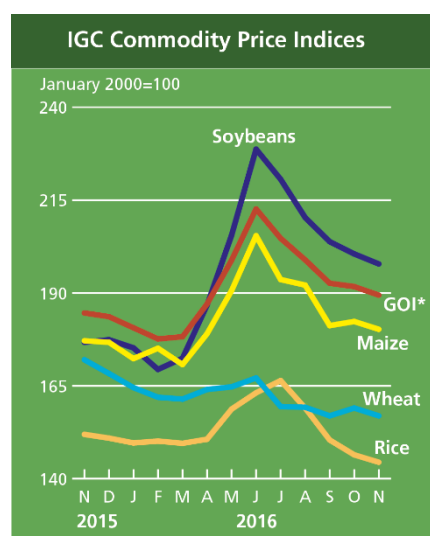
year. Old crop values in Argentina were firmer m/m, underpinned by slow country movement.

Rice

Amid continued pressure from sluggish global demand and advancing main crop harvests in Asian exporters, the IGC GOI rice sub-Index eased by around 1 percent in November, with values touching nine-year lows more recently. However, quotations in Viet Nam were underpinned by rain delays to the summer-autumn harvest, while hopes for fresh sales to the Philippines were mildly supportive. FOB prices in Pakistan were firmer on tight supplies and concerns about the milling quality of new crop arrivals. Outside of Asia, broken rice values at the US Gulf were little changed m/m.

Soybeans

Average global soybean values weakened for the fifth consecutive month during November, the IGC GOI sub-Index down by about 1 percent m/m. International demand for US supplies provided underlying support throughout the month, with 2016/17 export commitments around one-quarter higher y/y and equivalent to more than 70 percent of the full marketing year forecast. However, this was outweighed by pressure from a heavy fundamental backdrop; in addition to the completion of a record US harvest, seeding of 2016/17 crops advanced quickly in South America, notably in Brazil, under generally beneficial conditions. External markets were also influential at times, with US dollar strength adding to the bearish tone.

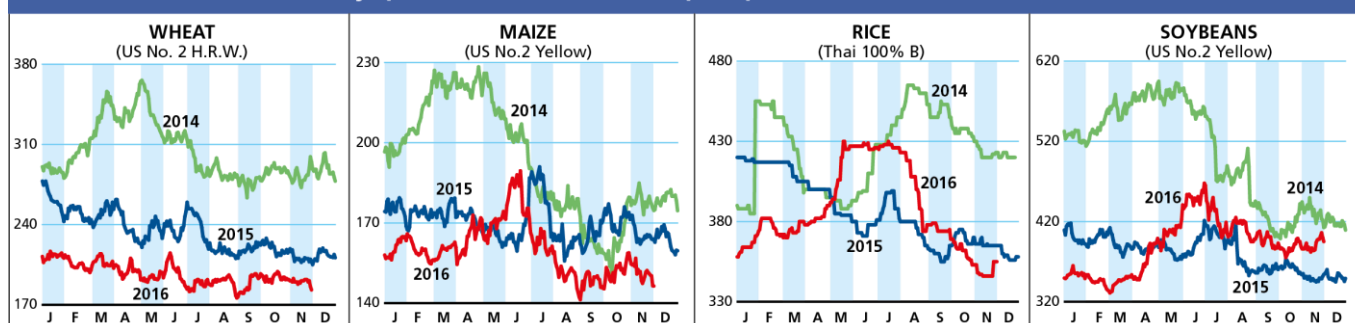


*GOI: Grains and Oilseeds Index

		IGC commodity price indices				
		GOI*	Wheat	Maize	Rice	Soybeans
(..... January 2000 = 100) (.....)						
2015	November	184.6	172.0	177.1	151.9	176.7
	December	183.6	168.3	176.7	150.9	177.4
2016	January	180.6	164.5	172.3	149.6	175.2
	February	177.6	161.9	175.1	150.1	169.4
	March	178.2	161.4	170.7	149.5	172.4
	April	187.0	164.0	179.0	150.6	186.7
	May	198.8	164.7	190.6	158.7	205.5
	June	212.6	167.1	205.4	163.1	228.7
	July	204.7	159.4	193.6	166.4	220.6
	August	198.9	159.2	192.1	159.0	210.3
	September	192.6	156.9	181.2	150.4	203.8
	October	191.7	159.0	182.3	146.4	200.5
	November	189.4	156.9	180.2	144.4	197.8

Selected export prices, currencies and indices

Daily quotations of selected export prices (USD/tonne, 2014-2016)

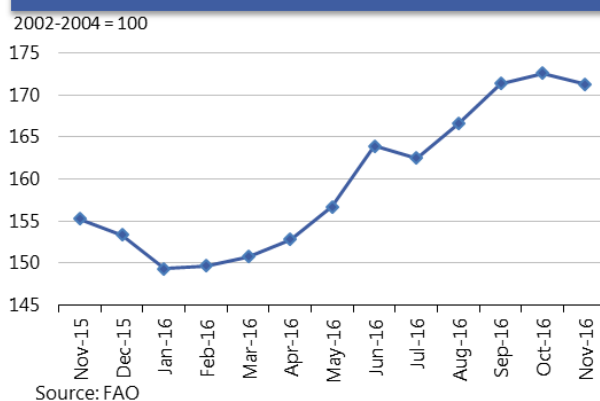
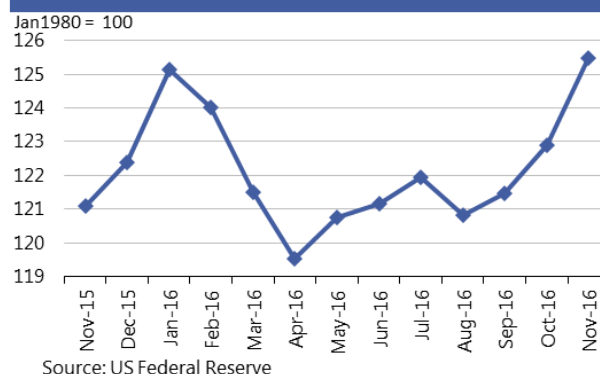


Daily quotations of selected export prices

	Effective Date	Quotation (1)	Week ago (2)	Month ago (3)	Year ago (4)	% change (1) over (2)	% change (1) over (4)
(..... USD/tonne) (.....)							
Wheat (US No. 2, HRW)	02-Dec	180	190	192	205	-5.3%	-12.2%
Maize (US No. 2, Yellow)	02-Dec	146	151	155	163	-2.7%	-10.0%
Rice (Thai 100% B)	02-Dec	364	355	353	365	2.5%	-0.3%
Soybeans (US No.2, Yellow)	02-Dec	393	402	386	355	-2.2%	10.7%

AMIS Countries' Currencies Against US Dollar

AMIS Countries	Currency	November 2016 Average	Monthly Change	Annual Change
Argentina	ARS	15.31	-0.96%	-59.44%
Australia	AUD	1.33	-1.11%	5.07%
Brazil	BRL	3.34	-4.92%	11.53%
Canada	CAD	1.34	-1.49%	-1.28%
China	CNY	6.84	-1.68%	-7.40%
Egypt	EGP	15.98	-79.90%	-102.30%
EU	EUR	0.93	-2.21%	0.52%
India	INR	67.67	-1.42%	-2.35%
Indonesia	IDR	13,311.45	-2.27%	2.64%
Japan	JPY	108.54	-4.53%	11.48%
Kazakhstan	KZT	339.49	-2.25%	-11.82%
Rep. Korea	KRW	1,162.92	-3.21%	-0.88%
Mexico	MXN	20.08	-6.36%	-20.80%
Nigeria	NGN	304.22	0.22%	-52.98%
Philippines	PHP	49.20	-1.83%	-4.57%
Russian Fed.	RUB	64.45	-3.03%	0.78%
Saudi Arabia	SAR	3.75	0.00%	0.02%
South Africa	ZAR	13.94	-0.15%	1.45%
Thailand	THB	35.35	-0.83%	1.20%
Turkey	TRY	3.29	-7.10%	-14.52%
UK	GBP	0.80	0.88%	-22.11%
Ukraine	UAH	25.68	0.06%	-10.98%
Viet Nam	VND	22,430.86	-0.70%	-0.07%

FAO Food Price Index
Nov2015 - Nov2016Nominal Broad Dollar Index
Nov2015 - Nov2016

Futures markets

Futures Prices – nearby

	Nov-16 Average	% Change	
		M/M	Y/Y
Wheat	148	- 1.4%	-18.5%
Maize	136	- 1.2%	- 5.8%
Rice	211	- 5.8%	- 20.2%
Soybeans	368	+2.8%	+15.4%

Source: CME

Futures prices

Prices for wheat and maize drifted about 1 percent lower m/m, with little fresh news arising from either supply or demand side to impact prices. Soybeans rose 2.8 percent, buoyed by export demand. Additionally, an expansion of biodiesel program announced in the 2017 EPA renewable fuel standards mandate, caused a late month price surge in soy oil, positively impacting soybeans. Rice prices, weighed by the ample US domestic and global supply situation, slumped to multi-year lows midmonth, reflecting a 6 percent m/m decline. A strong USD kept prices on the defensive, particularly for wheat which faces the greatest competition in the export market. Wheat, maize and rice prices were down 18, 6 and 20 percent y/y respectively, while soybean prices were up 15 percent.

Volumes and volatility

Wheat and maize volumes rose m/m by over 50 percent while soybean volumes dropped by 33 percent, tracking normal seasonal patterns for all three commodities. Both historical and implied volatility declined modestly for all three commodities reflecting subdued values ranging from 16 to 25.

Basis levels and transport

With the maize and soybean harvest glut easing, basis levels rose slightly in the interior. In Illinois the interior bids to local elevators were minus USD 6 and minus USD 11 per tonne the nearby futures for maize and soybeans respectively, while in Iowa the bids for the respective crops were minus USD 15 and minus USD 28. Gulf export market, however, was weak. Maize and soybeans were quoted at multi-year seasonal lows of USD 13 and USD 11 over respective futures, partly attributable to a sharp drop in barge freight m/m and a mid-month surge in barge loadings from the upper Mississippi before the winter closure of the locks. SRW export values were steady at about 18 per tonne over the nearby futures. Notably, a tow of 42 grain barges ran aground on the lower Mississippi (Memphis), underscoring low water levels and future necessity to lighten loads without rain relief. Export commitments of about 42.3 million tonnes continued to outpace last year's levels of 28.1 million tonnes as did cumulative shipments (48.4 vs. 36.0 million tonnes) for the respective marketing years of wheat, maize and soybeans.

Historical Volatility – 30 Days, nearby

	Monthly Averages		
	Nov-16	Oct-16	Nov-15
Wheat	25.5	29.5	27.0
Maize	23.0	24.7	17.0
Rice	26.9	22.3	25.7
Soybeans	16.4	18.4	14.6

Forward curves

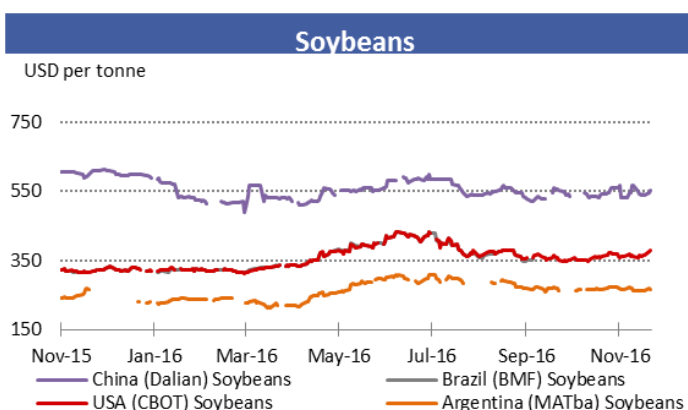
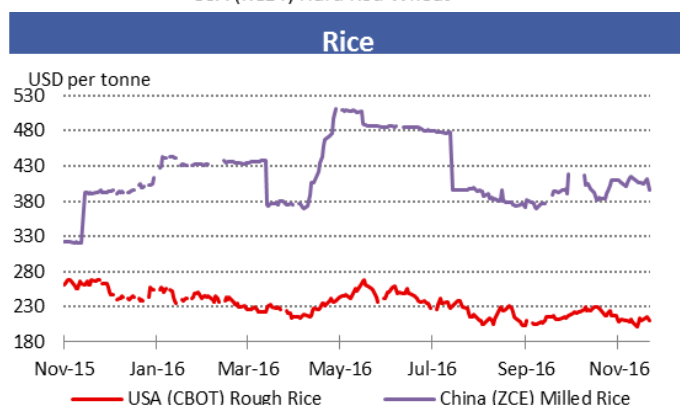
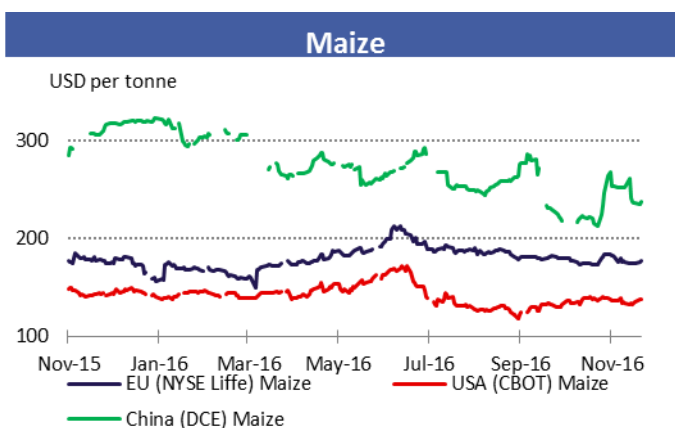
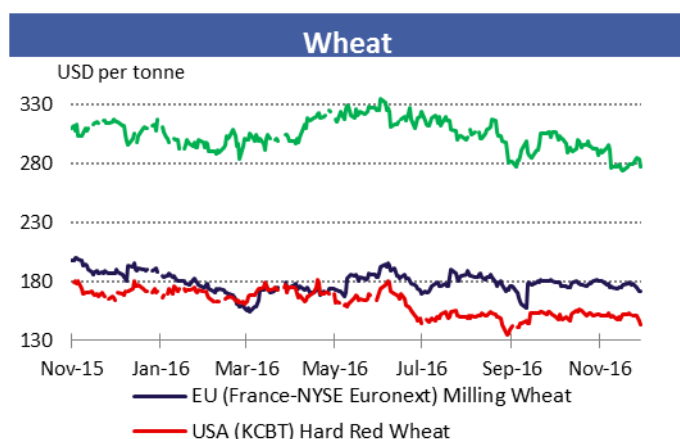
Forward curves for wheat, maize and soybeans exhibited unchanged configurations compared to last month. Wheat and maize maintained their wide upward sloping configuration (contango), while soybean curves continued to exhibit a small inversion of approximately USD 5 between old crop (January) and new crop (November, 2017). Despite low basis levels, no deliveries were issued on the December maize futures while 900 contracts (122,400 tonnes) were tendered on the December wheat futures (soybeans are not deliverable in December).

Investment flows

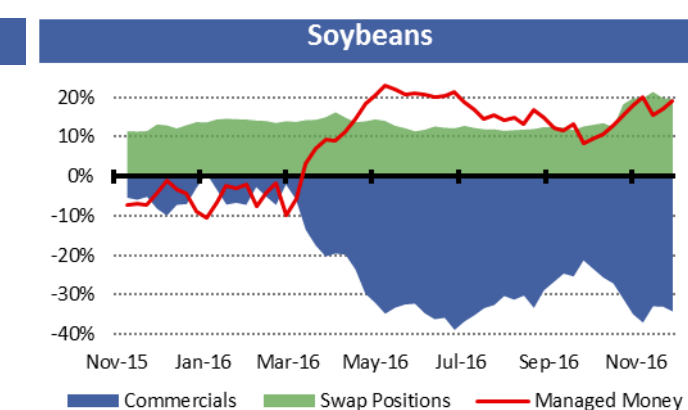
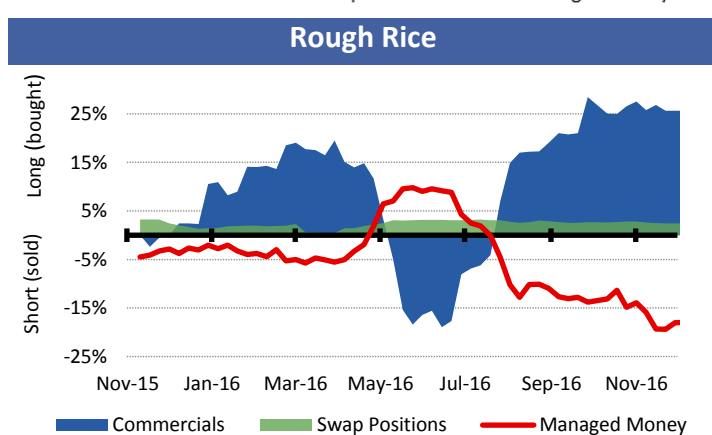
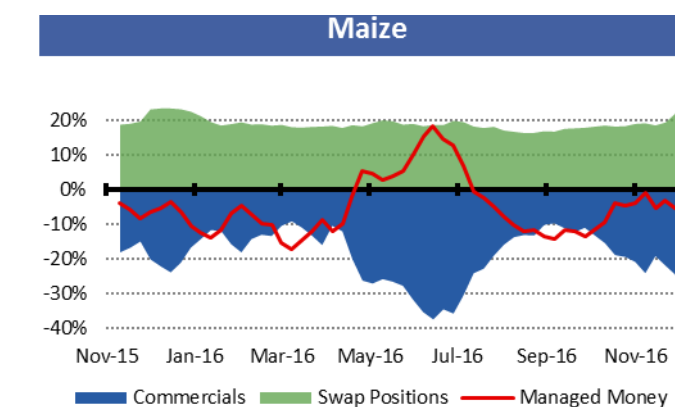
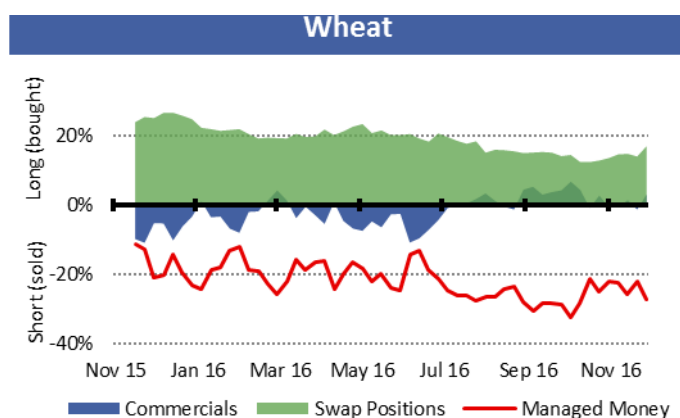
Managed money maintained nearly identical positions m/m, holding on to net short positions in wheat and maize and a net long position in soybeans m/m. Commercials also tended to stay the course on the opposite side. The only activity of note occurred within the swaps dealers category which increased its net long position in soybeans by about 10,000 contracts m/m. As these positions are established by small retail investors, they are normally observed as contrary indicators to future price direction.

Market indicators

Daily quotations from leading exchanges - nearby futures

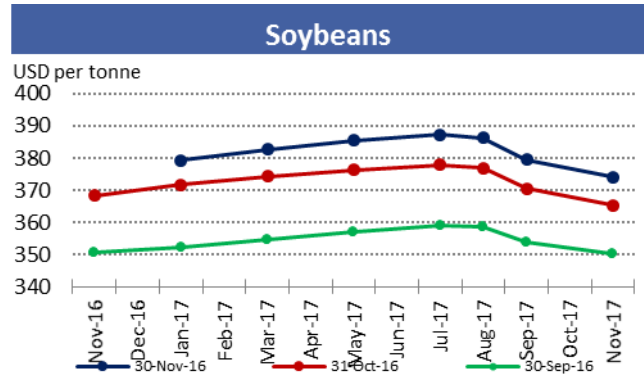
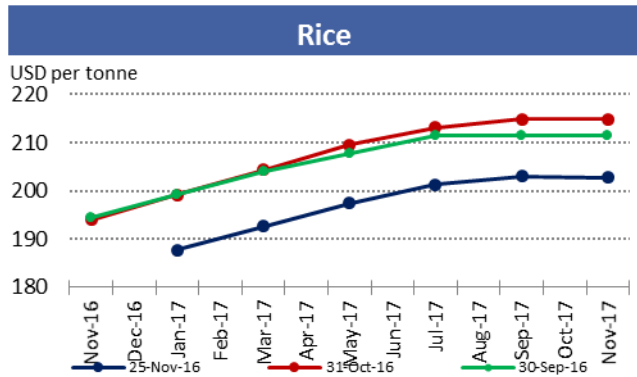
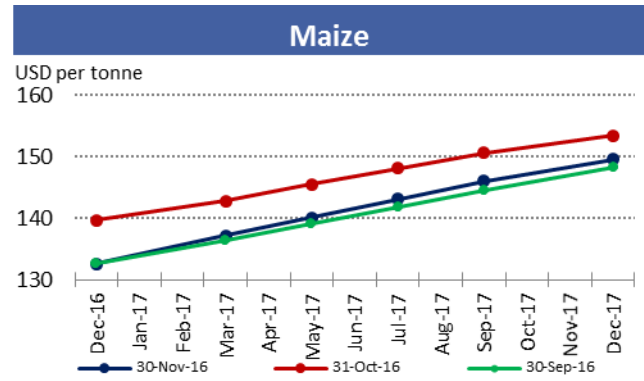
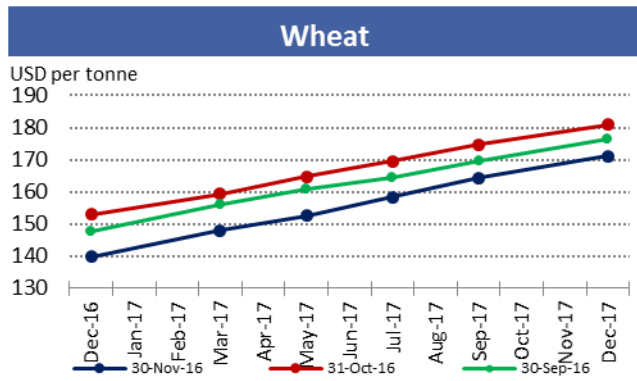


CFTC Commitments of Traders - Major Categories Net Length as percentage of Open Interest*

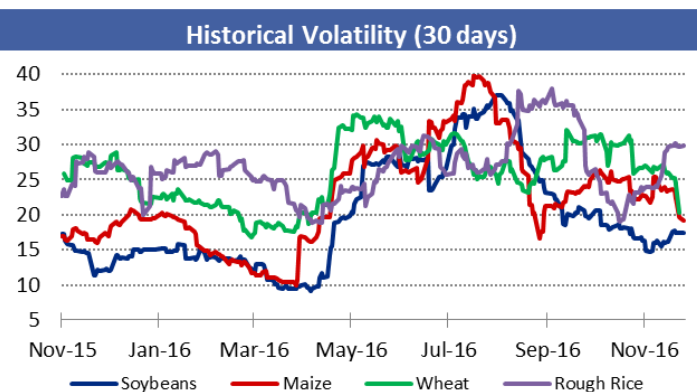
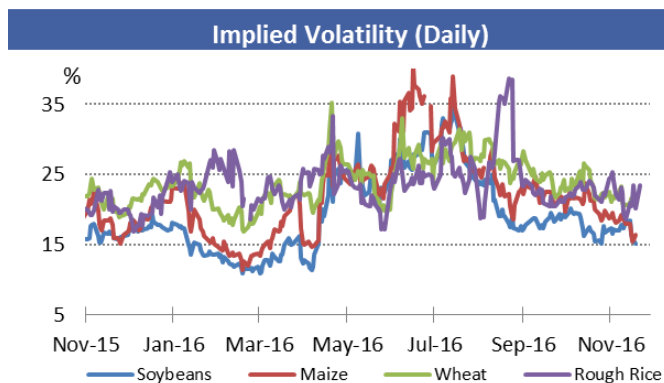


*Disaggregated Futures Only. Though not all positions are reflected in the charts, total long positions always equal total short positions.

Forward Curves



Historical and Implied Volatilities



AMIS Market indicators

Some of the indicators covered in this report are updated regularly on the AMIS website. These, as well as other market indicators, can be found at: <http://www.amis-outlook.org/amis-monitoring/indicators/>

For more information on technical terms please view the Glossary at the following link:

http://www.amis-outlook.org/fileadmin/user_upload/amis/docs/Market_monitor/Glossary.pdf

Monthly US ethanol update

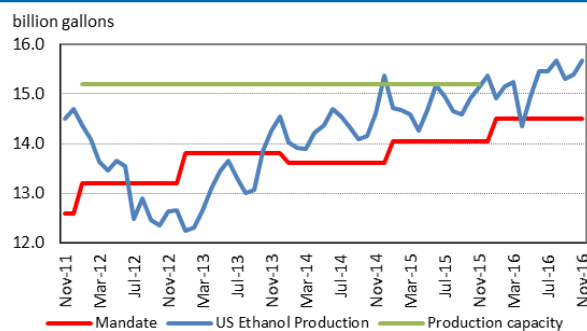
- **Ethanol production margins** were unchanged from the prior month but well above year-ago-levels, changes in input and output prices were offsetting.
- **DDGs prices** continued to weaken relative to maize, falling below 90 percent of the price of maize in the last full week in November.
- **Ethanol futures** continue to trade at a premium to RBOB gasoline, with the premium strengthening over last month to a ratio consistent with year-ago levels. The ratio strenghtend in the second half of November.
- On 23 November the US Environmental Protection Agency (EPA) released its final **Renewable Fuel Standard** for 2017 and biodiesel standard for 2018.
- The final 2017 volumes showed revisions from the preliminary rule released earlier in the year that included expanding the conventional gap, the potential space for maize ethanol, to 15 billion gallons. This reflects the maximum volume outlined in the Energy Independence and Security Act where the volumes were outlined.
- The final rule for volumes in 2017 also increased the advanced portion of the mandate compared to the preliminary rule, and a reading of the text of the final rule reveals an EPA expectation that some of the advanced volume, above that specifically allocated to biodiesel, will be met by additional volumes of biodiesel.

Spot prices	Nov 2016*	Oct 2016	Nov 2015
IA, NE and IL/eastern corn belt average			
Maize price (USD per tonne)	127.29	126.55	139.70
DDGs (USD per tonne)	107.15	109.88	124.41
Ethanol price (USD per gallon)	1.56	1.55	1.43
Nearby futures prices			
CME, NYSE			
Ethanol (USD per gallon)	1.55	1.57	1.49
RBOB Gasoline (USD per gallon)	1.4	1.5	1.3
Ethanol/RBOB price ratio	112.0%	105.6%	112.1%
Ethanol margins			
IA, NE and IL/eastern corn belt average, USD per gallon)			
Ethanol receipts	1.56	1.55	1.43
DDGs receipts	0.33	0.34	0.38
Maize costs	1.18	1.17	1.29
Other costs	0.55	0.55	0.55
Production margin	0.17	0.16	-0.02
Ethanol production			
(million gallons)			
Monthly production total	1,287	1,306	1,43
Annualized production pace	15,664	15,379	15,123

Based on USDA data and private sources

* Estimated using available weekly data to date.

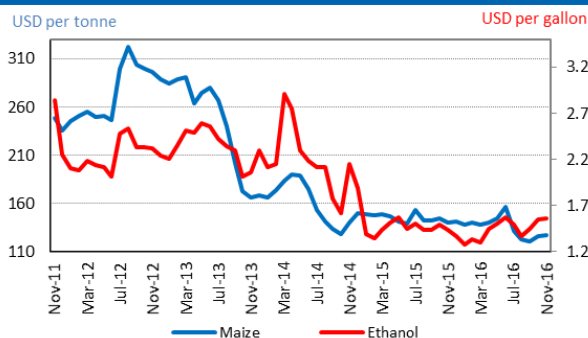
Ethanol production pace, capacity and annual mandate



Ethanol Production Margin
(IA, NE, IL/eastern corn belt average)



Ethanol price vs. maize price
(Spot prices)



Ethanol and RBOB gasoline
(nearby futures prices, CME, NYSE)

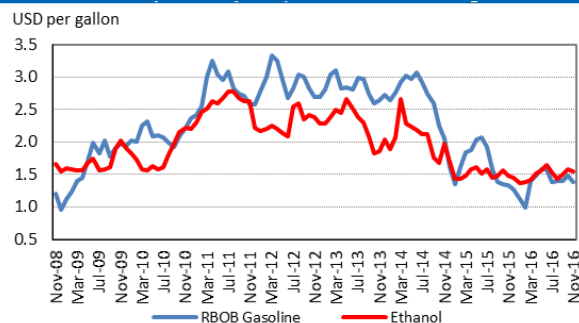


Chart and tables description

Ethanol Production Margins: The ethanol margin gives an indication of the profitability of maize-based ethanol production in the United States. It uses current market prices for maize, Dried Distillers Grains (DDGs) and ethanol, with an additional USD 0.55 per gallon of production costs

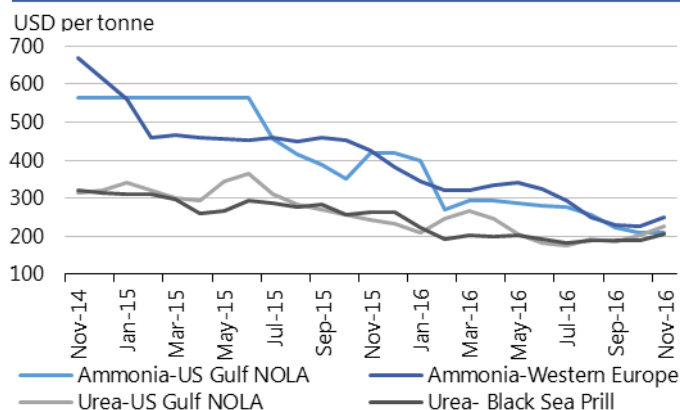
Ethanol Production Pace, Capacity and Mandate: Overview of the volume of maize-based ethanol production in the United States; it also highlights overall production capacity and the production volume that is mandated by public legislation. Name-plate (i.e. nominal) ethanol production capacity in the US is roughly 14.9 billion gallons per annum, but plants can exceed this level, so the actual capacity is assumed to be 15.2 billion gallons.

DDGs: By-product of maize-based biofuel production, commonly used as feedstuff.

RBOB: Reformulated Blendstock for Oxygenate Blending, gasoline nearby futures (NYSE).

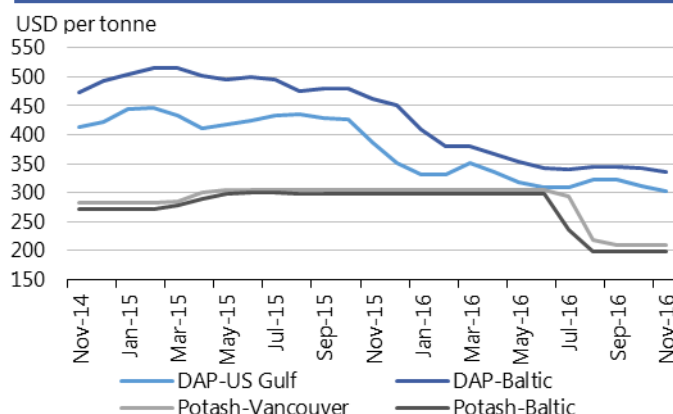
Fertilizer outlook

Ammonia and Urea (Spot prices)



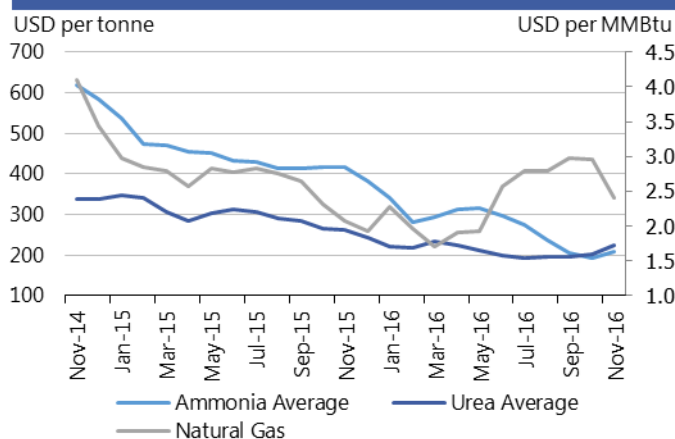
Note: calculations based on Bloomberg.

Potash and Phosphate (Spot prices)



Note: calculations based on Bloomberg.

Ammonia Average, Urea Average and Natural Gas (Spot prices)



Note: Natural gas is used as major input to produce nitrogen-based fertilizers. Own elaboration based on Bloomberg.

- The average prices of **nitrogen-based fertilizers** increased following an important rise in coal prices, China's key nitrogen cost driver
- **Ammonia** prices remained weak in the US after low application rates during the fall. In contrast, the price increase in Western Europe suggests a market recovery in the region.
- The average price of **urea** increased following China's decision to hold its supplies. However, demand may be affected by India's abrupt decision to cancel purchases; this decision was mainly due to the government's resolution to ban high denomination bills to deter illegal transactions. Additionally, weaker currencies in several emerging markets after the US elections have pushed prices upward.
- **DAP** m/m prices in both the US Gulf and Baltic hit the lowest level in 12 months. Weak demand from Indian buyers and high competition are pushing prices down.
- **Potash** m/m prices held steady despite production cuts by Canadian producers to reduce their inventories.
- The m/m price of **natural gas** continued to follow a downward trend as demand remained low due to higher-than-normal temperatures in the US.

Region	November average	November std. dev	% change last month	% change last year	12-month high	12-month low
Ammonia-US Gulf NOLA	210.0	-	-	-0.5	565.0	210.0
Ammonia-Western Europe	250.0	16.8	0.11	-0.4	667.5	225.0
Urea-US Gulf	224.5	15.4	0.12	-0.1	363.6	174.4
Urea-Black Sea	205.5	8.7	0.09	-0.2	320.0	183.0
DAP-US Gulf	301.5	2.4	-0.03	-0.2	447.0	301.5
DAP-Baltic	335.0	-	-0.02	-0.3	515.0	335.0
Potash-Baltic	198.0	-	-	-0.3	300.0	198.0
Potash-Vancouver	209.0	-	-	-0.3	305.0	209.0
Ammonia Average	208.6	12.3	-0.09	-0.5	617.8	191.3
Urea Average	224.4	10.7	0.1	-0.1	347.6	192.8
Natural Gas	2.4	0.2	-0.19	0.2	4.1	1.7

Source: Bloomberg



Chart and tables description

Ammonia and Urea: Overview of nitrogen-based fertilizer prices in the US Gulf, Western Europe and Black Sea. Prices are weekly prices averaged by month.
Potash and Phosphate: Overview of phosphate and potassium-based fertilizer prices in the US Gulf, Baltic and Vancouver. Prices are weekly prices averaged by month.
Ammonia Average and Urea Average: Monthly average prices from Ammonia's US Gulf NOLA, Middle East, Black Sea and Western Europe were averaged to obtain Ammonia Average prices; monthly average prices from Urea's US Gulf NOLA, US Gulf Prill, Middle East Prill, Black Sea Prill and Mediterranean were averaged to obtain Urea Average prices.
Natural Gas: Henry Hub Natural Gas Spot Price from ICE. Prices are intraday prices averaged by month. Natural gas is used as major input to produce nitrogen-based fertilizers. **DAP:** Diammonium Phosphate.

Explanatory Notes

The notions of **tightening** and **easing** used in the summary table of “**World Supply and Demand**” reflect judgmental views which take into account market fundamentals, inter-alia price developments and short-term trends in demand and supply, especially changes in stocks.

All totals (aggregates) are computed from unrounded data. World supply and demand estimates/forecasts in this report are based on the latest data published by FAO, IGC and USDA; for the former, they also take into account information received from AMIS countries (hence the notion “**FAO-AMIS**”). World estimates and forecasts may vary due to several reasons. Apart from different release dates, the three main sources may apply different methodologies to construct the elements of the balances. Specifically:

Production: For wheat, production data refer to the first year of the marketing season shown (e.g. the 2014 production is allocated to the 2014/15 marketing season). For maize and rice, FAO-AMIS production data refer to the season corresponding to the first year shown, as for wheat. However, in the case of rice, 2014 production also includes secondary crops gathered in 2015. By contrast, for rice and maize, USDA and IGC aggregate production of the northern hemisphere of the first year (e.g. 2014) with production of the southern hemisphere of the second year (2015 production) in the corresponding 2014/15 global marketing season. For soybeans, this latter method is used by all three sources.

Supply: Defined as production plus opening stocks. No major differences across sources.

Utilization: For wheat, maize and rice, utilization includes food, feed and other uses (“other uses” comprise seeds, industrial utilization and post-harvest losses). For soybeans, it comprises crush, food and other uses. No major differences across sources.

Trade: Data refer to exports. For wheat and maize, trade is reported on a July/June marketing year basis, except for the USDA maize trade estimates, which are reported on an October/September basis. For rice, trade covers flows from January to December of the second year shown, and for soybeans from October to September. Trade between European Union member states is excluded.

Stocks: In general, stocks refer to the sum of carry-overs at the close of each country’s national marketing year. In the case of maize and rice, in southern hemisphere countries the definition of the national marketing year is not the same across the three sources as it depends on the methodology chosen to allocate production. For Soybeans, the USDA world stock level is based on an aggregate of stock levels as of 31 August for all countries, coinciding with the end of the US marketing season. By contrast, the IGC and FAO-AMIS measure of world stocks is the sum of carry-overs at the close of each country’s national marketing year.

Main sources

Bloomberg, CFTC, CME Group, FAO, GEOGLAM, IFPRI, IGC, Reuters, USDA, US Federal Reserve

AMIS - GEOGLAM Crop Calendar

Selected leading producers

Wheat		J	F	M	A	M	J	J	A	S	O	N	D
EU (21%)*	winter												
China (17%)	spring												
	winter												
India (13%)	winter												
US (8%)	spring												
	winter												
Russia (8%)	spring												
	winter												
Maize		J	F	M	A	M	J	J	A	S	O	N	D
US (35%)													
China (22%)	north												
	south												
Brazil (8%)	1st crop												
	2nd crop												
EU (7%)													
Argentina (3%)													
Rice		J	F	M	A	M	J	J	A	S	O	N	D
China (29%)	intermediary crop												
	late crop												
	early crop												
India (21%)	kharif												
	rabi												
Indonesia (9%)	main Java												
	second Java												
Viet Nam (6%)	winter-spring												
	summer/autumn												
	winter												
Thailand (4%)	main season												
	second season												
Soybeans		J	F	M	A	M	J	J	A	S	O	N	D
USA (31%)													
Brazil (29%)													
Argentina (18%)													
China (4%)													
India (3%)													

* Percentages refer to the global share of production (average 2013-15).

	Planting (peak)		Harvest (peak)
	Planting		Harvest
	Weather conditions in this period are critical for yields.		Growing period

2016 AMIS Market Monitor Release Dates

04 February, 03 March, 07 April, 05 May, 02 June, 07 July, 08 September, 06 October, 10 November, 08 December

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